Minkyoo Song

Mail | LinkedIn | Google Scholar CV Site

last update: September 2026

Research Interest

LLM Security, AI for Security, Data-driven Security, Social Network Analysis

Education

- **Korea Advanced Institute of Science and Technology (KAIST)** *March 2023 - February 2027 (estimated)*
  Ph.D. Candidate in Electrical Engineering, Network and System Security Lab (Advisor: Seungwon Shin) Daejeon, South Korea
- **Korea Advanced Institute of Science and Technology (KAIST)** *March 2021 - February 2023*
  M.S. in Electrical Engineering, Network and System Security Lab (Advisor: Seungwon Shin) Daejeon, South Korea
- **Korea Advanced Institute of Science and Technology (KAIST)** *March 2016 - February 2021*
  B.S. in Industrial and Systems Engineering, double majored in Electrical Engineering Daejeon, South Korea

Publications [C]: Conference, [J]: Journal, [U]: Under Review

[C] Y. Kim, **M. Song**, S. Shin. **SkillMutator: Benchmarking and Defending Language-and-Code Cross-modal Attacks on LLM Agent Skills**. *2026 Annual Computer Security Applications Conference (**ACSAC 2026**)*

[C] W. Choi*, M. Seo*, **M. Song**, H. Heo, S. Shin, M. You. **PC^2: Politically Controversial Content Generation via Jailbreaking Attacks on GPT-based Text-to-Image Models**. *33rd ACM Conference on Computer and Communications Security (**CCS 2026**)*

[C] J. Kim, **M. Song**, S. Shin, S. Son. **SafeMoE: Safe Fine-Tuning for MoE LLMs by Aligning Harmful Input Routing**. *The Fourteenth International Conference on Learning Representations (**ICLR 2026**)*

[C] J. Kim, S.H. Na, **M. Song**, S. Shin, S. Son. **MoEvil: Poisoning Expert to Compromise the Safety of Mixture-of-Experts LLMs**. *2025 Annual Computer Security Applications Conference (**ACSAC 2025**)* ***(Distinguished Paper Award)***

[C] **M. Song**, H. Kim, J. Kim, S. Shin, S. Son. **Refusal Is Not an Option: Unlearning Safety Alignment of Large Language Models**. *34th USENIX Security Symposium (**SECURITY 2025**)*

[C] H. Kim, **M. Song**, S.H. Na, S. Shin, K. Lee. **When LLMs Go Online: The Emerging Threat of Web-Enabled LLMs**. *34th USENIX Security Symposium (**SECURITY 2025**)*

[C] **M. Song**, H. Kim, J. Kim, Y. Jin, S. Shin. **Claim-Guided Textual Backdoor Attack for Practical Applications**. *The 2025 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (**NAACL 2025 Findings**)*

[C] J. Kim, **M. Song**, S.H. Na, S. Shin. **Obliviate: Neutralizing Task-Agnostic Backdoors within the Parameter-Efficient Fine-Tuning Paradigm**. *The 2025 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (**NAACL 2025 Findings**)*

[C] **M. Song**, E. Jang, J. Kim, S. Shin. **Covering Cracks in Content Moderation: Delexicalized Distant Supervision for Illicit Drug Jargon Detection**. *31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2025**)*

[C] J. Kim, **M. Song**, M. Seo, Y. Jin, S. Shin. **PassREfinder: Credential Stuffing Risk Prediction by Representing Password Reuse between Websites on a Graph**. *2024 IEEE Symposium on Security and Privacy (SP) (**Oakland 2024**)*

[J] J. Kim, **M. Song**, M. Seo, Y. Jin, S. Shin, J. Kim. **PassREfinder-FL: Privacy-Preserving Credential Stuffing Risk Prediction via Graph-Based Federated Learning for Representing Password Reuse between Websites**. *Elsevier Expert Systems with Applications (ESWA)*

[J] J. Choi, J. Kim, **M. Song**, H. Kim, N. Park, M. Seo, Y. Jin, S. Shin. **A Large-Scale Bitcoin Abuse Measurement and Clustering Analysis Utilizing Public Reports**. *IEICE Transactions on Information and Systems*

[U] **M. Song**, J. Kim, M. Kang, H. Kim, S. Shin, S. Son. **Graphs Don't Stay Secret: Practical Subgraph Reconstruction Attacks on Defended Graph RAG**. *Under Review*

[U] K. Kim, J. Cui, **M. Song**, S. Shin. **Exploring the Familiar Taste of Toxicity: A Causal Influence Analysis of Toxic Comments on Internet Forums**. *Invited to Major Revision at IEEE Transactions on Knowledge and Data Engineering (TKDE)*

[U] H. Kim, J.Cui, **M. Song**, H. Heo, S. Shin, K. Lee, X. Liao. **SCRIPTIOC-BENCH: A Benchmark for Recognizing Actionable Threat Intelligence from Script-Based Malware using LLMs**. *Under Review*

[U] K. Kim, S.H. Na, **M. Song**, S. Shin. **Global Meta-path-level Counterfactual Explanation for Heterogeneous Graph Neural Networks by Path Exclusion**. *Under Review*

[U] J. Kim, M. Seo, **M. Song**, S. Shin, J. Kim. **To Make Each Account Count: Exploring Credential Data Breach Threats through Victim-driven Analysis**. *Under Review*

EXPERIENCE

- **Johns Hopkins University** *September 2026 - February 2027*
  Visiting Scholar Baltimore, MD, USA
  ◦ Working with Prof. Yinzhi Cao.
- **S2W [🌐]** *July 2022 - Feb 2023*
  Research Intern @ AI Team South Korea
  ◦ **Illicit drug jargon detection**: Analyzed illicit drug-related discussions and developed an LLM-based content moderation framework, independently capturing contextual and lexical characteristics.

HONORS AND AWARDS

- **Distinguished Paper Award** *2025*
  Annual Computer Security Applications Conference (ACSAC)
  ◦ MoEvil: Poisoning Expert to Compromise the Safety of Mixture-of-Experts LLMs
- **4th Prize, 2025 Cybersecurity Paper Competition** *2025*
  Korean Association of Cybersecurity Studies (KACS)
  ◦ Poisoning Expert to Compromise the Safety of Mixture-of-Experts LLMs
- **2nd Prize, 2023 Cybersecurity Paper Competition** *2023*
  Korean Association of Cybersecurity Studies (KACS)
  ◦ Graph-based Deep Learning Framework for Credential Stuffing Risk Prediction
- **4th Prize, 2023 Cybersecurity Paper Competition** *2023*
  Korean Association of Cybersecurity Studies (KACS)
  ◦ Delexicalized Distant Supervision for Illicit Drug Jargon Detection
- **4th Prize, 2023 Cybersecurity Paper Competition** *2023*
  Korean Association of Cybersecurity Studies (KACS)
  ◦ Understanding the Occurrence and Impact of Credential Data Breach
- **Cum Laude** *2021*
  Korea Advanced Institute of Science and Technology (KAIST)
- **Academic Achievement Award: Salutatorian** *2019 Spring*
  Korea Advanced Institute of Science and Technology (KAIST)
- **Dean's List** *2019 Spring*
  Industrial and Systems Engineering (ISysE, KAIST)

LANGUAGES

Korean (Native), English (Fluent)